3c3 Analogue circuits MOSFET Amplifiers Lab

By Attila Garganego (22339628)

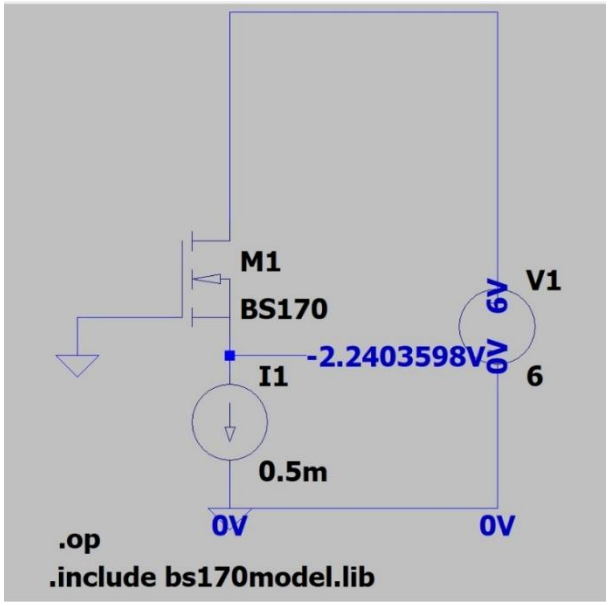
Variables chosen for I_d, V_{ds} and amplification

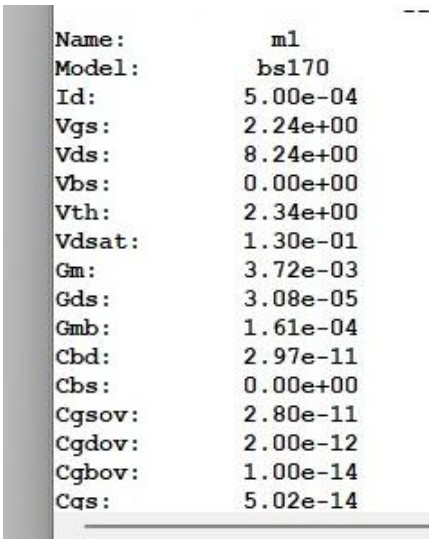
The variables chosen were:

Amplification	I_d	V_{ds}
2	0.5mA	8V

Gm and r0

The gm and r0 are calculated using LTspice simulations





Name:	m1
Model:	bs170
Id:	5.00e-04
Vgs:	2.24e+00
Vds:	8.24e+00
Vbs:	0.00e+00
Vth:	2.34e+00
Vdsat:	1.30e-01
Gm:	3.72e-03
Gds:	3.08e-05
Gmb:	1.61e-04
Cbd:	2.97e-11
Cbs:	0.00e+00
Cgsov:	2.80e-11
Cgdov:	2.00e-12
Cgbov:	1.00e-14
Cgs:	5.02e-14

To calculate r0 this formula was used:

$$r0 = \frac{1}{G_{ds}}$$

r0	32468 Ω
Gm	0.00372 S

Finding R_s, R_d & V_{dd}

Simplified formula for amplitude

$$|A_v| = g_m(R_d||R_l)$$

Assuming R_l will be equivalent to R_d the value chosen was 27kΩ. The simplified value calculated had an amplitude value of 50. This is way beyond our wanted amplitude, but the r0 value is not

much bigger than the Rd values this means that the simplified formula diverges strongly from the giveb Av.

Formula used to calculate V_{dd}

$$V_{dd} = (R_d + R_s) * I_d + V_{ds}$$

The resistance value of Rd was based on Rl being the same value was chosen arbitrarily with some importance on outing most of the voltage on Rd and making sure that Vdd didn't surpass 30V.

Resistances	Current	Voltages
Rd = 27kΩ	0.5mA	13.5V
Rs = 6.8kΩ	0.5mA	3.4V

V_{dd} = 13.5 + 8 + 3.4 = 24.9V ≈*25.6V

*Due to a calculation error the Vdd used is 25.6 instead of 24.9 volts

Gate resistance calculations

The voltage formula for V_{g2}

$$V_{g2} = V_s + V_t$$

V_s	3.4V
V_t	2.1V
V_{g2}	5.5V

The V_{g1} formula

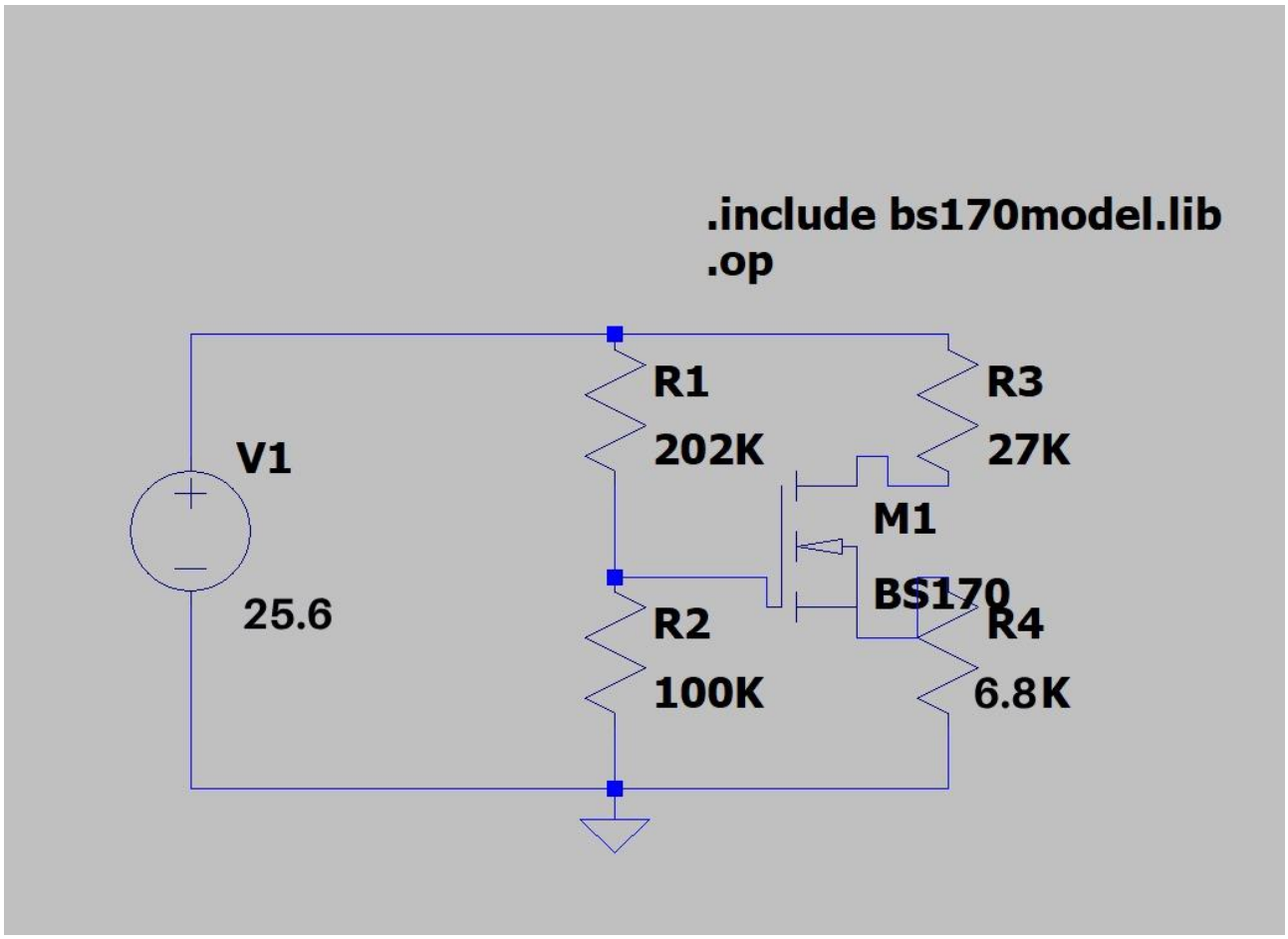
$$V_{g1} = V_{dd} - V_{g2}$$

V_{g2}	5.5V
V_{dd}	25.6V
V_{g1}	20.1V

Ratio

Real Ratio	Simplified Ratio	Resistance ratio	Approximate Ratio
20.1 : 5.5	3.654 : 1	229.5kΩ : 62.8kΩ	232kΩ : 62.8kΩ

The values were then tested using LTspice simulation to make sure the Gm values gives us the proper Id value of 0.5mA



The I_d value remained the same 0.4991 mA

Capacitor choice

The effective range of this amplifier frequency wise is 100hz-10000hz

The capacitor equation to calculate impedance

$$X_I = \frac{1}{2\pi C f}$$

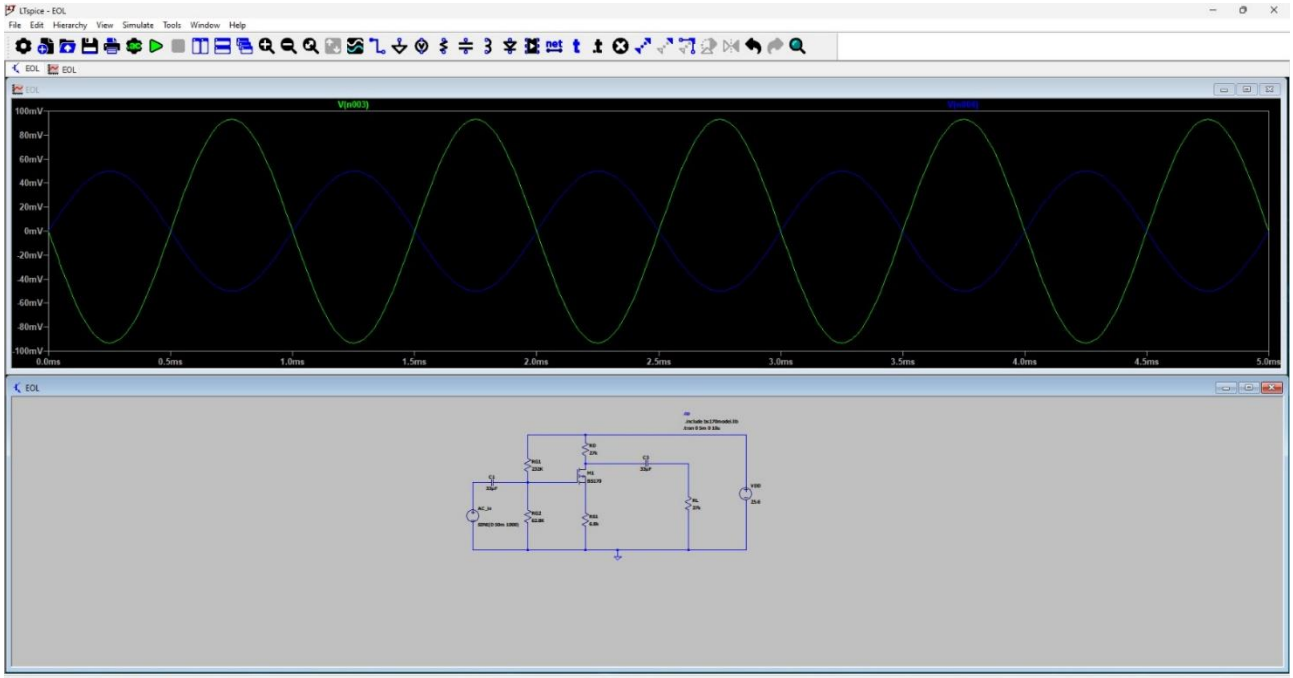
Capacitor	Impedance
33 µF	50Ω

The capacitors chosen where 33µF. This was because they gave an effective impedance to the lowest frequency of 50 ohms. Which is much smaller than the kilo-ohms of resistance values.

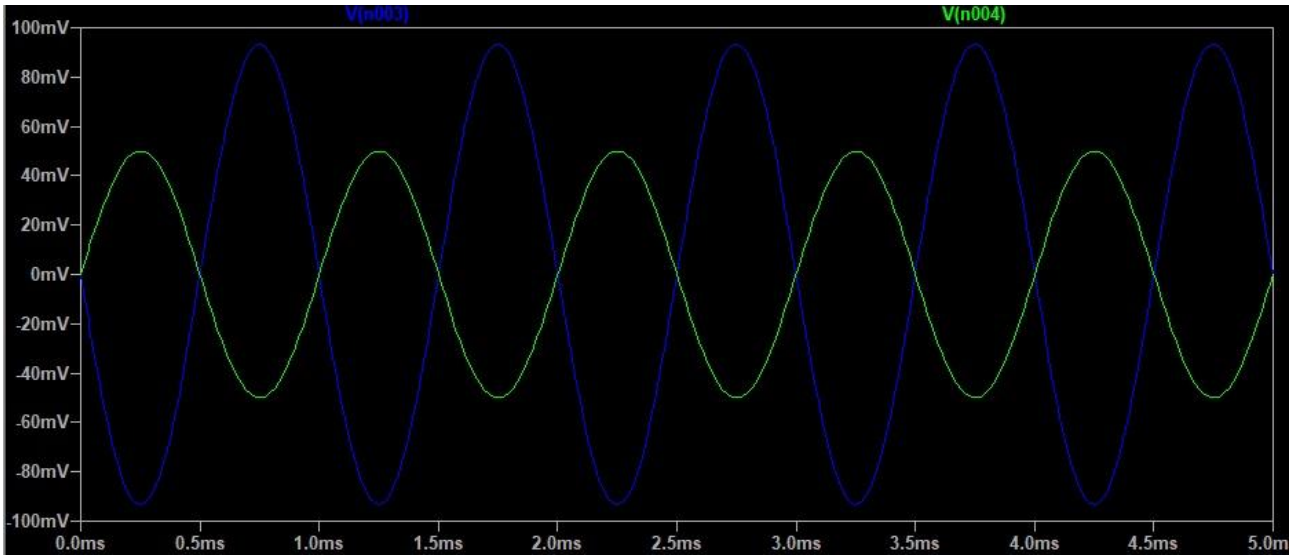
Theoretical Amplitude

$$A_v = \frac{-g_m(R_d||R_L)}{1 + g_m * R_s + \frac{R_d||R_L + R_s}{r_0}}$$

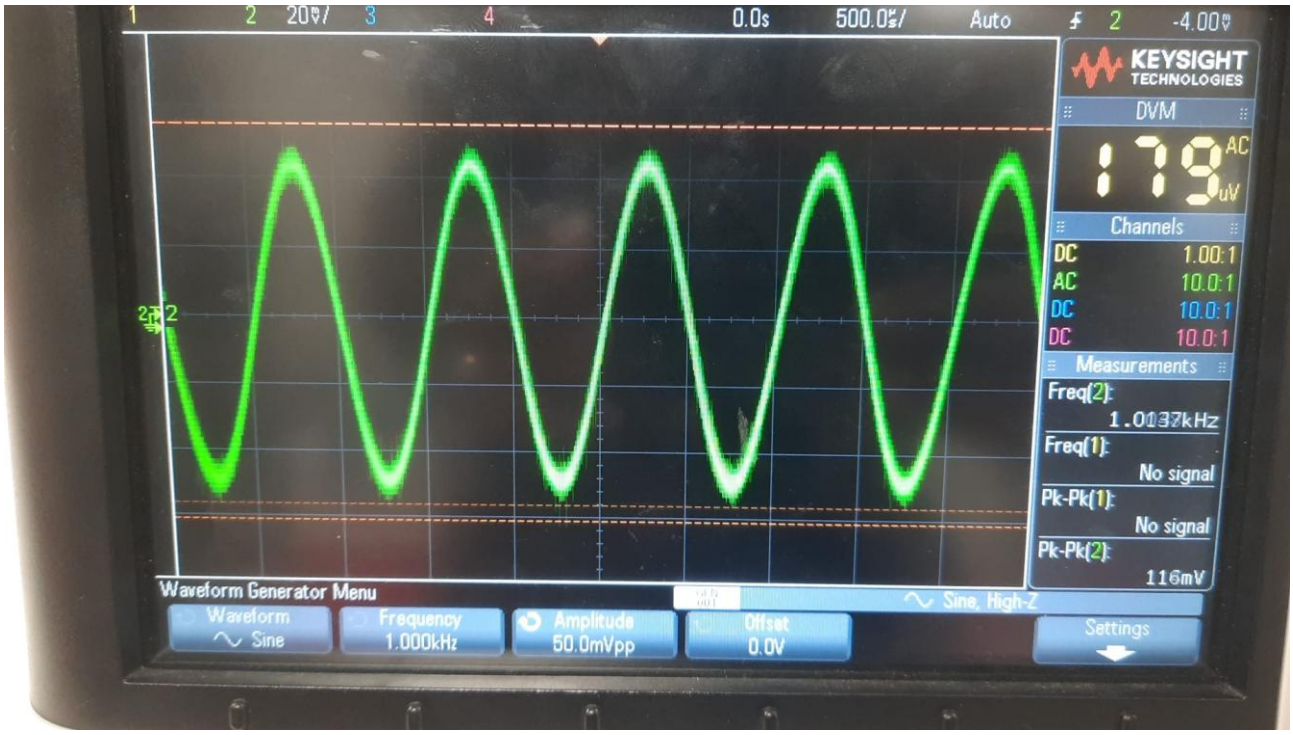
gm	Rd	Rl	r0	Rs	Av
3.72*10-3 S	27kΩ	27kΩ	32468Ω	6.8kΩ	-1.865

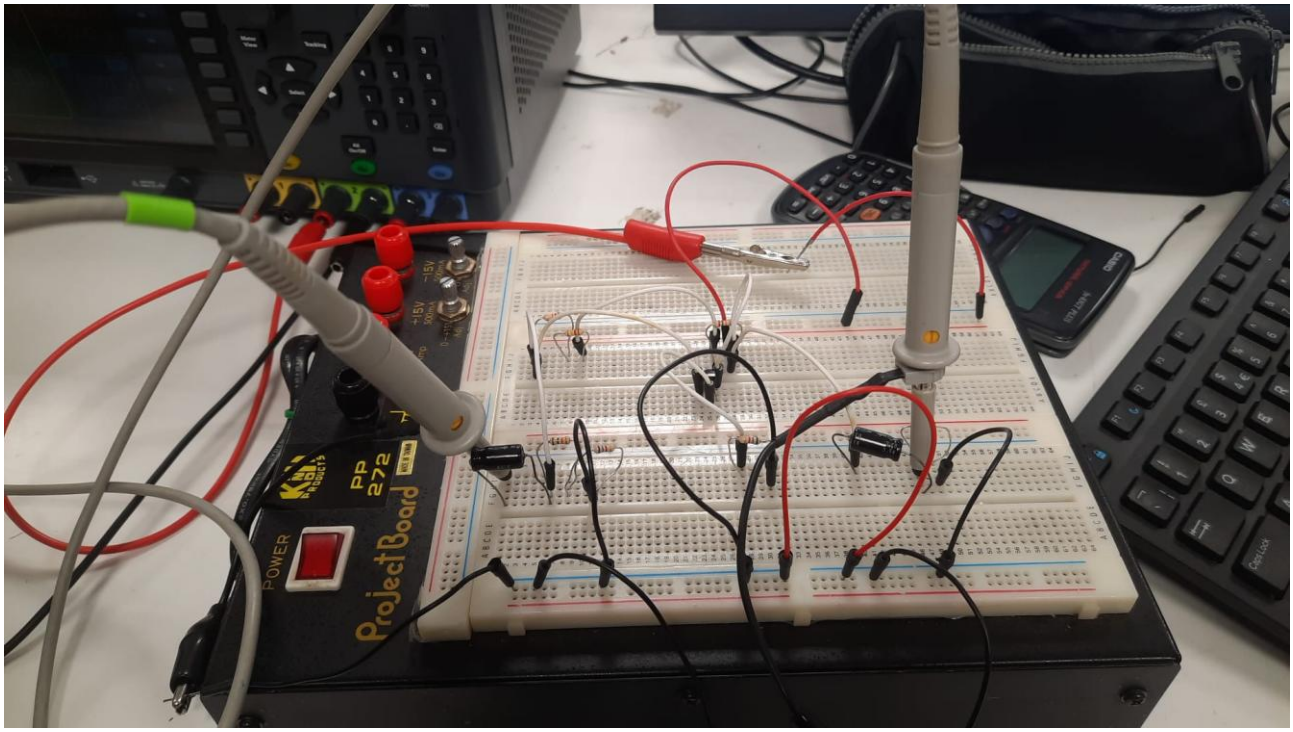


x = 0.747ms y = 93.30mV



Real Circuit waveform and circuit





Discussion

Using the wrong voltage value, gave me the wrong amplification value, from 1.8 to 2. So it has worked, but unintentionally. This was all caused by me calculating incorrectly the voltage values when adding them together.